

The Gate Learns Hygiene: The Training Economics of the Write Path in Linear-RNN Language Models

Gunner Levi Howe
gunnerlevihowe@gmail.com

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Abstract

Linear-RNN language models write into a fixed-size state on every token, and their two best-known capability gaps — associative recall and state tracking — are both plausibly downstream of undisciplined writes. The frontier now ships learned write gates (Gated DeltaNet-2), but no one has reported what a learned write gate actually *learns*, or whether write discipline causally buys either capability during training. We pre-registered and ran a 13-run controlled grid at 30.0M parameters (300M tokens per run): gate arms {always-write, L_1 -priced learned gate, unpriced learned gate, oracle gate} $\times$ tracking-data mixture {0, 2, 10, 50%}, on TinyStories plus synthetic state-tracking families with token-level role annotations held out from every learned arm. Three results. (1) *The price discovers hygiene*: with a small L_1 price on writing, the learned gate ends training at mean 0.1187 on state-operation tokens versus 0.4728 on content tokens — recovering the oracle’s write-suppression pattern with no access to role labels — while the unpriced gate stays $2.8\times$ more open on operations; the gap is present at every mixture and grows with operation density. These are, to our knowledge, the first token-role statistics of a learned write gate. (2) *Hygiene buys recall, cheaply but weakly*: on pure language data the priced-gate arm beats always-write on multi-query recall by $+0.033/+0.022$ across two seeds at matched perplexity (worst-case cost $+0.5\%$) — directionally replicated, $2\text{--}3\times$ an identical-configuration noise floor we measure directly, but below our pre-registered $+0.03$ margin at one seed, and from a near-chance base. (3) *Hygiene does not buy automata*: the pre-registered gating-helps-tracking-under-scarcity prediction died with the sign inverted, the only tracking skill that emerged anywhere arose in the *always-write* arm, and damping op-token writes at inference *destroys* that skill ($1.00 \rightarrow 0.5$) rather than revealing anything beneath — inverting the reveal effect our companion mechanism study established in cornered toy training, and completing a three-regime scope map: write damping reveals hidden automata in minimal-width pure training, repairs them in hybrid tasks, and deletes skills in LM training, where the write path is how skills form. All predictions, kills, and analysis rules were pre-registered in-repository before any grid run; deviations are disclosed, and every number in this paper regenerates from committed run artifacts.

1 Introduction

A linear-RNN language model carries its entire context in a fixed-size matrix state, updated by a write on every token. Its two most-cited capability gaps relative to attention — associative recall of stated facts [1] and multi-step state tracking [8, 4] — have a shared candidate root: *undisciplined writing*. Every token overwrites state whether or not it carries information worth storing, so stored facts get polluted (the recall gap), and state-transformation structure gets buried (the tracking gap).

The field has begun to act on this intuition at the architecture level. Gated DeltaNet-2 [6] decouples erase and write with channel-wise learned gates at 1.3B parameters and reports better recall and perplexity; sparse-write designs improve recall in expanded-state settings [9, 14]; the lineage goes back through gated delta rules [16, 15, 5] to learned update-skipping in classical RNNs [2]. But shipping a gate is not understanding one. To our knowledge, no published work reports *what a learned write gate learns* — its statistics against token roles — nor whether write discipline *causally* buys recall or tracking during training, nor what it

costs. Those are economics questions, and they are design-rule-relevant precisely because the frontier is already paying for the machinery.

This paper asks the economics questions at garage scale, with pre-registered discipline. We train 30.0M-parameter DeltaNet-class language models from scratch under four write-gate arms — **A** always-write ($g_t \equiv 1$, the field default), **B** learned gate with an L_1 price on writing, **B'** learned gate unpriced (the Gated DeltaNet-2 species), and **C** oracle gate (writes suppressed on ground-truth state-operation tokens) — crossed with a data-mixture axis $\{0, 2, 10, 50\%\}$ of synthetic state-tracking text whose token roles (content / operation / check) are annotated in the corpus but visible only to arm C. The synthetic families sit at three algebraic tiers (memory-permutation S_3 , involution $\mathbb{Z}_2$, cyclic $\mathbb{Z}_3$), with evaluation chains up to $4\times$ deeper than training. Every prediction and kill criterion was frozen in-repository before the first grid run.

Contributions.

1. **The first token-role account of a learned write gate** (§4.1). Under an L_1 price, the gate discovers write hygiene unsupervised: mean gate 0.1187 on operation tokens vs. 0.4728 on content at the 50% mixture (gap +0.3541), with 35.5% of gates effectively shut. The unpriced control gates operations at 0.3346 — $2.8\times$ more open — and barely sparsifies (7.0% shut). Exploratory: the hygiene gap is dose-responsive in mixture (+0.197 $\rightarrow$ +0.2772 $\rightarrow$ +0.3541).
2. **A measured recall effect with honest error bars** (§4.2). Priced hygiene lifts recall over always-write on pure LM data in both seeds run (+0.033, +0.022; matched perplexity), $2\text{--}3\times$ the identical-configuration noise floor (0.011) that our own grid supplies, but below the pre-registered +0.03 margin at seed 1 and from a near-chance base — reported as supported and fragile, not confirmed.
3. **A clean pre-registered negative on the tracking side** (§4.3–§4.5). Gating did not help tracking emerge anywhere on the grid; at 2% scarcity the oracle arm was *worse* than always-write (predicted $> +0.10$, observed -0.07). The one skill that emerged — perfect depth- ≤ 8 toggle tracking, always-write arm, 50% mixture — lives *on* the write path: damping op-token writes at inference deletes it (1.00 $\rightarrow$ 0.5 at depth 8) while doing exactly nothing (0.0) where no skill exists. With our companion mechanism study [7], this completes a three-regime scope map for write-path intervention (reveal / repair / destroy) whose boundary variable is how cornered training is.
4. **Method disclosures the field can reuse**: an identical-computation noise floor hiding inside any oracle-mask grid (§4.2); a constant-policy artifact that makes at-chance accuracy tuples repeat across independent runs (§4.3); and a pre-registered, frozen-threshold emergence-forecasting protocol whose gate correctly declined to fire on this grid (§4.6).

Everything here is 30.0M-parameter scale on TinyStories-class data: we are validating and killing *design rules*, not making frontier claims. The rules that survive are candidates for the next scale up; the rule that died, died cheaply.

2 Related work

Write gating in linear attention. Gated DeltaNet [16] added a scalar decay gate to the delta rule [15]; Gated DeltaNet-2 [6] decouples erase from write with independent channel-wise sigmoid gates and reports recall and perplexity gains at 1.3B — establishing the machinery but reporting no gate behavioral statistics and no causal isolation of the write side. Skip-RNN [2] learned binary state-update skipping in classical RNNs with a budget penalty — the closest historical relative of our priced gate — but predates linear-attention states and reports skip rates, not token-role alignment. Sparse-write memories [9, 14] route writes to state partitions for recall; none prices writing or measures learned-gate semantics.

State tracking in linear RNNs. Extended-eigenvalue and Householder-product designs give DeltaNet-class models the expressivity for hard state tracking [4, 11]; trained models still fail to length-generalize, and recent work explicitly frames the gap as optimization, not expressivity [12, 10, 13]. Siems et al. [12] show always-write DeltaNet learns S_5 perfectly from pure synthetic REPL traces — fixing the undiluted

regime as solved and moving the open question to *diluted* streams, which motivated our mixture axis. Our companion study [7] gives a causal mechanism at toy scale: the additive input path is a parasitic attractor that conceals learned automata, and damping it at inference restores them. Whether that reveal effect survives real LM training is exactly this paper’s P4.

Recall evaluation. We use MQAR-style multi-query associative recall [1] rendered in our corpus format, TinyStories [3] as the base LM distribution, and report all chance levels inline.

3 Setup

Model. A 30.0M-parameter DeltaNet-class LM ($d=512$, 8 layers, 8 heads, tied embeddings, 8k BPE): per head, state $S \in \mathbb{R}^{d_k \times d_v}$ updates by the delta rule with $\beta_t \in [0, 2]$ (negative-eigenvalue range [4]) and a scalar per-head write gate g_t on the *value write only* (erase and routing untouched): $S_t = S_{t-1}(I - \beta_t k_t k_t^\top) + g_t \beta_t v_t k_t^\top$. Training uses a chunk-parallel WY formulation (pure PyTorch, chunk 128) verified equivalent to a sequential reference to 2×10^{-5} before any run. Runs are 24400 steps $\times$ 12288 tokens = 300M tokens, AdamW, bf16, one RTX 3080; evaluation every 2500 steps with per-eval checkpoints retained.

Arms. **A:** $g_t \equiv 1$. **B:** $g_t = \sigma(w_g^\top x_t)$ plus loss term $\lambda |g|$, $\lambda=3 \times 10^{-3}$ frozen from a pilot sweep. **B’:** same learned gate, $\lambda=0$ (the Gated DeltaNet-2 species, scalar rather than channel-wise). **C:** g_t set by corpus annotations — writes suppressed on operation tokens of synthetic segments, free elsewhere; the causal upper bound on hygiene. Arms A/B/B’ never see annotations.

Data. TinyStories mixed with synthetic tracking text at mixture $m \in \{0, 2, 10, 50\}\%$: three families rendered as natural sentences — *boxes* (3-object permutation tracking, S_3 tier), *toggle* (one lamp, $\mathbb{Z}_2$), *dial* (mod-3 counter, $\mathbb{Z}_3$) — with per-step state statements making tracking directly loss-bearing, training chain depths 1–8, and token-level role spans (content/operation/check) recorded in the shards. A generator self-test re-simulates 1,000 documents from raw text before any run.

Evals and chance. Tracking: held-out fixed-depth chains at depths $\{2, 4, 8, 16, 32\}$, $n=50$ per family $\times$ depth cell (binomial 95% half-width ≈ 0.14 at $p=0.5$; candidate-restricted scoring; chance = 0.5 toggle, 1/3 dial and boxes). Recall: MQAR [1] with 4/8/12 key–value pairs ($n=60$ each; aggregate chance 0.153). LM quality: TinyStories validation perplexity. Gate behavior: mean g_t and fraction of gates < 0.1 , plus token-role statistics against the held-out oracle masks.

Pre-registration. Predictions P1–P6 and kills K1–K3 were frozen in-repository before any grid run (commit b868a75); two amendments (batch-size/heartbeat infrastructure after a silent-VRAM-spill false start; additive forecasting instrumentation) were committed before any result existed. The frozen seed rule: any cell deciding a *positive* verdict is replicated at a second seed before the verdict is claimed; negatives stand on seed 0. The low-mixture (2%) cells were additionally held blind behind a separate forecasting freeze (§4.6) and opened only after that gate’s verdict was committed. Deviations from the frozen plan are disclosed in §6.

4 Results

4.1 P2: the price discovers hygiene

The pre-registered headline bet: a small L_1 price on writing makes the learned gate (i) sparse and (ii) *role-aligned* — quieter on state-operation tokens than on content — without ever seeing role labels. Both halves confirmed (Fig. 1a). At the 50% mixture the priced gate ends training at mean 0.3384 with 35.5% of gates below 0.1; measured against the held-out oracle masks, it averages 0.1187 on operation tokens vs. 0.4728 on content/other tokens (gap +0.3541; 64 training-format sequences, 10058 operation tokens). The unpriced control B’ — same architecture, same data, $\lambda=0$ — ends at mean 0.4999 with only 7.0% shut and operation-token gate 0.3346: gating exists, hygiene is weak. The price is what buys the role alignment, at a perplexity cost never exceeding +0.5% (P6 passes; K3 never fires).

Two robustness notes. First, gate behavior replicates: at $m=0$ the priced gate’s final mean is 0.5404 vs. 0.543 across two seeds. Second (exploratory, not pre-registered): the hygiene gap exists at *every* mixture and grows with operation density — +0.197 at 2%, +0.2772 at 10%, +0.3541 at 50% — a dose-response pattern consistent with the gate learning from the statistics of the operation tokens themselves.

arm	mix%	ppl	toggle-8	dial-8	boxes-16/32	MQAR	gate mean	gates off
A	50	3.35	1.00	0.34	0.38	0.161	1.000	0%
B	50	3.36	0.50	0.34	0.37	0.150	0.338	36%
C	50	3.32	0.50	0.32	0.34	0.200	0.644	36%
B'	50	3.35	0.50	0.34	0.34	0.122	0.500	7%
A	10	3.12	0.50	0.32	0.34	0.189	1.000	0%
B	10	3.12	0.50	0.34	0.29	0.150	0.519	4%
C	10	3.13	0.50	0.32	0.41	0.206	0.975	3%
A	2	3.08	0.50	0.32	0.40	0.144	1.000	0%
B	2	3.09	0.50	0.34	0.42	0.161	0.535	2%
C	2	3.10	0.50	0.32	0.33	0.144	1.000	0%
A	0	3.08	0.50	0.32	0.31	0.161	1.000	0%
B	0	3.06	0.50	0.34	0.35	0.194	0.540	2%
C	0	3.08	0.50	0.32	0.34	0.150	1.000	0%

Table 1: Final evaluation, all 13 grid runs (seed 0; chance: toggle 0.5, dial/boxes 1/3, MQAR 0.153). The grid’s three headline facts: only always-write at 50% mixture masters toggle (1.00 vs. 0.50 everywhere else); dial-8 is at chance in every run trained on dial data; and at the 50% mixture the priced gate (B) is the only *learned* arm whose gate sparsifies (36% shut vs. 7% unpriced), at unchanged perplexity.

To our knowledge these are the first published token-role statistics of a learned write gate in any linear-attention model. The measurement is model-agnostic — it needs only gate activations and a role-annotated probe corpus — and applies directly to production gates (e.g. Gated DeltaNet-2’s), where whether trillion-token training discovers hygiene on its own is an open, now-testable question.

4.2 P3: hygiene buys recall — small, replicated in direction, honestly fragile

The two-gap link prediction: on pure language data ($m=0$), the priced-gate arm beats always-write on MQAR by $> +0.03$ at matched perplexity. Measured: $+0.033$ at seed 0 (pass) and $+0.022$ at seed 1 (below margin), both at slightly *better* perplexity (-0.6% , -0.3%), mean $+0.028$ (Fig. 1b). Under the frozen seed rule this is **supported directionally, below margin at one of two seeds** — a small, real-looking effect requiring more seeds, not a confirmed > 3 -point claim.

Three calibrations the grid itself provides. (i) *Noise floor*: at $m=0$ the oracle mask is all-ones, so arm C and arm A specify the same computation; their end-of-training difference — $|\Delta\text{MQAR}| = 0.011$, $|\Delta\text{track}| = 0.016$ — is a direct identical-configuration replicate floor (hardware nondeterminism plus one numerically-inert mask multiply). The P3 effect is $2\text{--}3\times$ this floor at both seeds. (ii) *Chance framing*: aggregate MQAR chance is 0.153; arm A sits essentially at it (0.1611/0.15), so the honest statement is that hygiene lifts recall from $\approx$ chance to slightly above (0.1944/0.1722), and unevenly across load at seed 0 (0.333 vs. 0.3 at 4 pairs, 0.1 vs. 0.117 at 8, 0.15 vs. 0.067 at 12) — absolute recall at this scale is weak for every arm. (iii) *Exploratory cross-check*: the *oracle* arm also out-recalls always-write where it actually gates ($m>0$: 0.2 vs. 0.1611 at 50%, 0.2056 vs. 0.1889 at 10%), same sign as P3, not pre-registered.

4.3 P1/P5: write discipline does not buy tracking — sign inverted at scarcity

The tracking-side hypothesis — write discipline helps tracking emerge, most where tracking data is rare — died by its own pre-registered criteria. P1 predicted oracle-over-always-write deep-boxes advantage $> +0.10$ at the 2% mixture; measured -0.07 — sign inverted. P5 predicted $C > A$ at every mixture; measured 0.07 at 10%, -0.04 at 50%, -0.07 at 2% — mixed signs within a few noise floors. The fuller truth visible in Table 1: *deep boxes never emerged in any arm* — all thirteen runs sit in a chance band around 1/3 — so the P1/P5 contrasts are movements inside noise on a task nobody learned. Gating didn’t rescue deep tracking; nothing did, at this scale and budget.

Two honest instrument notes. First, at-chance cells collapse to *constant answer policies*: a model that always answers “on” scores exactly the eval pack’s gold-label frequency at each depth, which is why identical

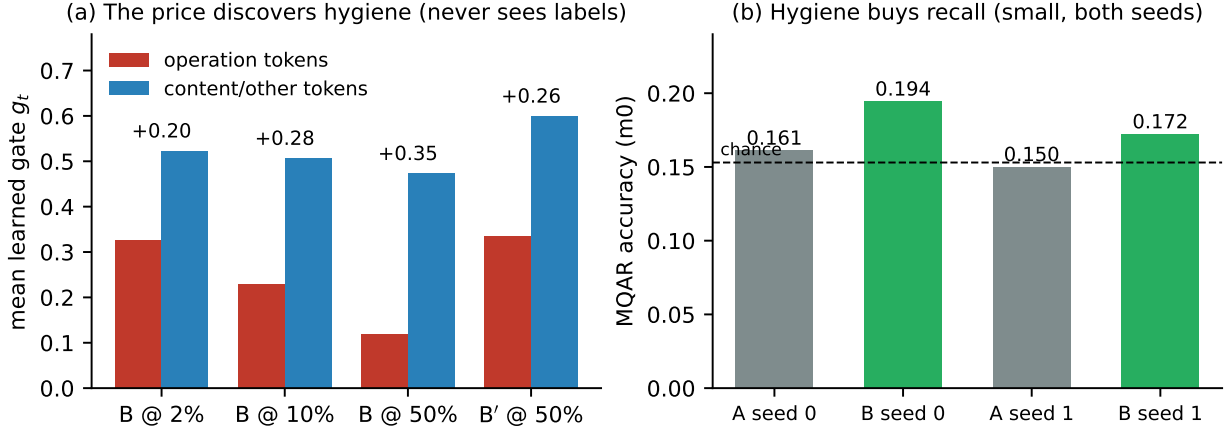


Figure 1: **(a)** Mean learned gate on operation vs. content tokens (held-out oracle masks). The priced gate (B) suppresses writes on operations at every mixture, hardest where operations are densest; the unpriced gate (B') shows the same sign but 2.8 $\times$ weaker suppression. **(b)** MQAR recall at $m=0$ (pure LM data), both seeds: priced hygiene lifts recall from a near-chance base at matched perplexity.

accuracy tuples recur across independent runs in Table 1 — repeated values are a fingerprint of at-chance behavior, not copied evaluations. Second, dial-8 sits in $[0.32, 0.34]$ (chance 0.333) in *every* run trained on dial data — the fourth consecutive experimental generation in this program (toy grids, 5-seed note grids, and now every LM arm) in which a cyclic mod-3 task drifts while same-tier non-cyclic structure locks. We leave the abelian-drift wall as a named open problem; our own earlier mechanistic hypothesis for it was killed in Howe [7].

4.4 The emergence event that did happen — and what it is not

Exactly one tracking skill emerged on the grid: the *always-write* arm at the 50% mixture reached perfect toggle accuracy at depths 2/4/8 (1.0/1.0/1.0), already ≥ 0.9 by the first evaluation (2.5k steps) and perfect at every later one except a single transient reversion at 10k steps, while every gated arm — priced, unpriced, oracle — stayed at chance (0.5) all the way to 300M tokens (Fig. 2). Write gating didn't merely fail to help the easiest family; every form of it *prevented* the one emergence the always-write baseline found instantly.

Honesty requires deflating the event itself: the skill is depth-bounded, not an automaton. Accuracy falls to 0.54/0.44 at depths 16/32 — inside the training envelope it is perfect, one step beyond it is chance. Even the grid's sole winner is pattern mastery within the trained depth range, consistent with the unexplored-states account of recurrent length generalization [10], and *not* the length-generalizing “automaton underneath” of the toy regime [7].

4.5 P4: damping destroys — the reveal effect inverts at LM scale

Our companion study established, in cornered toy training, that damping the additive input path at inference *reveals* a correctly-learned automaton hiding under a parasitic shortcut [7]. P4 pre-registered the same intervention here: damp op-token value-writes on trained always-write checkpoints; predict improvement at $m=50$ (where tracking data forced a skill) and nothing at $m=0$.

Measured (Fig. 3): the $m=0$ half held exactly (0.0 composite change — the intervention is a true no-op where no skill exists), and the $m=50$ half *inverted*: damping op-token writes destroys the toggle mastery — $1.0 \rightarrow 0.52$ at depth 2, $1.0 \rightarrow 0.54$ at depth 4, $1.0 \rightarrow 0.5$ at depth 8; composite -0.16 . There is nothing underneath to reveal: the skill *lives on* the write path.

Combined with the toy and hybrid results, the program's mechanism now has a complete, pre-registered scope map:

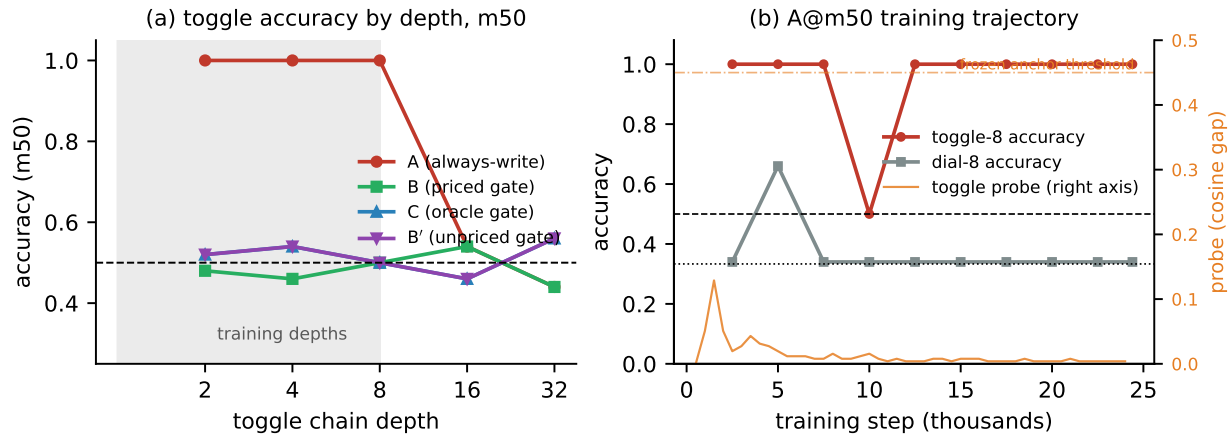


Figure 2: **(a)** Toggle accuracy by chain depth at the 50% mixture: always-write (A) is perfect across the training depths (shaded) and at chance beyond; every gated arm is at chance everywhere. **(b)** The A@m50 trajectory: toggle-8 is mastered by the first evaluation, with one transient reversion at 10k steps; dial-8 never leaves chance beyond one $n=50$ fluctuation; and the representation probe (right axis; within-minus-between class cosine gap at the answer position) stays weak throughout — maximum 0.1289, far below its frozen 0.45 anchor threshold — even while behavior is perfect.

training regime	write-path damping does	evidence
minimal-width, pure tasks (cornered)	reveals hidden automaton	Howe [7]
hybrid content+tracking tasks	repairs damaged solutions	Howe [7], hybrid bridge
LM training (this paper)	destroys the formed skill	P4, §4.5

The boundary variable is how cornered training is. When the loss admits only the automaton, writes are parasitic and damping strips the parasite. When training is a language-modeling problem, *writing is how the skill forms and where it lives*: the same intervention that exposes hidden structure in the toy deletes real structure in the LM. Any at-scale proposal built on the toy result’s “suppress the writes” intuition inherits this boundary — which is exactly why the cheap experiment was worth running before anyone runs the expensive one.

4.6 The pre-registered forecasting gate correctly declined to fire

This grid double-served as the test bed for a frozen emergence-forecasting protocol (dense probe plus behavioral anchors every 500 steps, composed-anchor event definition, thresholds (0.45, 0.65) frozen before any grid value was read, low-mixture cells held blind). Its own kill criterion fired: 0/7 calibration runs reached the frozen composite event bar — $\text{mean}(\text{toggle-8}, \text{dial-8}) \geq 0.9$ sustained — because the bar’s dial half never emerged anywhere (§4.3), and its toggle half, where it emerged at all, was already mastered by the first evaluation (1.0 toggle vs. 0.34 dial at $m=50$). A third, independent blocker: the representation-probe anchor itself carried no signal at this scale — its maximum over the entire A@m50 run was 0.1289, far below the frozen 0.45 firing threshold (Fig. 2b) — so the composed anchor could not have fired even had an approach window existed. Nothing was forecastable at this scale and cadence: one target never arrives, the other arrives before the first measurement, and the leading indicator that worked in the toy regime is mute here. The protocol’s verdict was committed *before* the blind cells were opened, no threshold was fitted post hoc, and any future relaxed-bar analysis of these artifacts is pre-labeled exploratory in the registration. We report this as the protocol working as designed — a frozen gate that declines to fire on inadequate data is the failure mode you want.

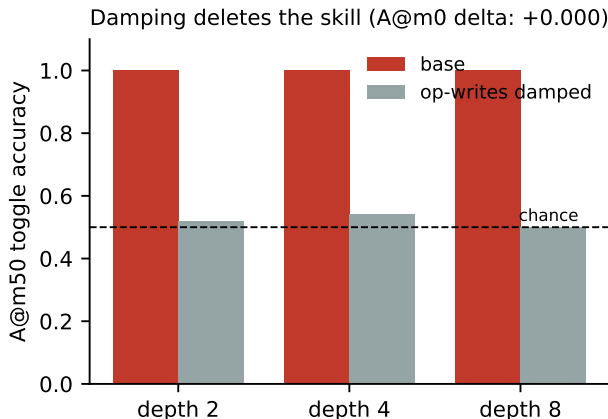


Figure 3: P4: inference-time damping of op-token value-writes on the trained A@m50 checkpoint deletes its toggle skill at every depth (composite -0.16); the same intervention on A@m0 changes nothing (0.0), confirming the probe is inert where no skill exists.

5 Discussion

For designers of gated linear-attention models. Three design-relevant facts, each pre-registered or disclosed exploratory. (1) A scalar L_1 price turns a write gate into a *hygiene* mechanism — role-aligned write suppression discovered from task statistics alone — at sub-1% perplexity cost; unpriced gates (the shipped default) leave most of that alignment on the table ($2.8\times$ weaker operation-suppression at matched architecture). (2) The payoff of hygiene, where we can measure it, is on the *memory* side (recall), not the *computation* side (tracking): priced hygiene lifted near-chance recall in both seeds at matched perplexity, while no form of write discipline — learned, priced, or oracle — helped any tracking family emerge. (3) Write freedom during skill formation matters: the only arm that mastered any tracking was the one allowed to write everywhere, and its skill demonstrably lives on those writes. A gate schedule that prices writes for state cleanliness *after* skills form is a different (untested) proposal than gating throughout training — our results caution against the latter and say nothing yet about the former.

The measurable question this hands the frontier. Our hygiene statistic — mean gate on operation vs. content tokens over a role-annotated probe corpus — transfers as-is to production checkpoints with learned write gates (Gated DeltaNet-2 class). Whether trillion-token training discovers hygiene without a price, and whether its gate statistics predict recall quality, are now concrete, cheap measurements on public checkpoints rather than speculation. That is the natural next experiment, and it requires no training at all.

The two-gap hypothesis, split. This program began from the intuition that recall pollution and tracking failure share the undisciplined-write root. The verdict is a clean split: the recall half survives (hygiene helps, a little, for free), the tracking half dies at LM scale (writes are load-bearing for skill formation), and the toy-regime reveal effect that motivated the tracking half is now scope-bounded to cornered training. “The gate learns hygiene; hygiene buys memory, not automata.”

6 Limitations and disclosures

Scale and power. 30.0M parameters, TinyStories-class data, S_3 -tier synthetic tracking, $n=50$ per tracking cell ($\pm \approx 0.14$), MQAR from a near-chance base, single seed except the pre-registered $m=0$ replication pair. The recall effect specifically needs more seeds before any $>\text{noise-floor-multiple}$ claim hardens. Gate parameterization is per-head scalar; Gated DeltaNet-2’s is channel-wise — our economics conclusions are parameterization-scoped.

Deviations from the frozen plan, disclosed. (i) The pre-registered $n_h=2$ extension pair was not run: the program’s decision questions resolved at $n_h=1$ and the budget moved to the seed replication. (ii) The frozen contingency (state-passing fine-tune [10] before any write-path attribution, triggered “if the

$m=50$ A arm fails deep tracking”) was not run: A@m50 mastered the toggle family P4 measures while deep *boxes* emerged in no arm (§4.3), and we judged the trigger’s intent — don’t attribute a *failure* to write behavior without the rescue control — inapplicable to P4’s actual finding, which attributes a *success* to the write path. Readers may weigh that judgment; the fine-tune remains the right control for any future deep-boxes attribution. (iii) Training shards were accidentally committed early in the program, silently blocking pushes; the unpushed range was rewritten to excise the data files (content otherwise unchanged), so the pre-registration commits carry third-party timestamps only from the day the block was found, with ordering enforced by the intra-repository hash chain. Old→new hashes are mapped in the repository’s VERDICTS.md. (iv) The role-annotated shards were deleted from disk in cleanup; the hygiene measurement (§4.1) runs on a deterministic regeneration of each training corpus prefix (seeded generators, frozen tokenizer), committed as `hygiene.json`; it reproduces the originally-logged values within sampling noise. (v) The forecasting protocol’s richer analyses (interval calibration, anchor-lead distributions) never activated because zero events occurred; only the gate verdict is claimed.

7 Reproducibility

All code, pre-registrations, run logs, verdict analysis, and this paper’s number/figure regeneration pipeline are published in the `twogap/` directory of <https://github.com/gunnerhowe/automaton-underneath>, alongside the companion mechanism study and its 202-run artifact set. The program’s working repository carries the commit chain that enforces the pre-registration ordering (private; access on request); key commits (post-excision hashes): pre-registration freeze `b868a75`; forecasting freeze `e84d431` and amendment `a260732`; forecasting verdict committed before blind-cell opening `f10b406`; grid verdicts `c4f2692`; P4 `11701e8`. Every number in this manuscript is a macro generated by `gen_numbers.py` from committed artifacts (`runs/*/log.json`, `hygiene.json`, `p4_results.json`, `fusion_constants.json`); `verify_regen.py` re-derives them and byte-compares before any release. Model checkpoints ($\approx 18\text{GB}$) are retained locally and available on request. Training is single-GPU (RTX 3080, $\approx 6\text{h/run}$); the full grid is ≈ 3.5 GPU-days.

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